Shadows of the Hall–Janko geometries

Human Compression project: bounded application

This note applies the accepted orbit and residue theorem to the complete rank-three and rank-four Hall–Janko catalogue. Its primary comparison is Jacobs and Leemans, *An algorithmic analysis of the intersection property*, LMS J. Comput. Math. 7 (2004), 284–299, Tables 1–4 and 7–8. The full common proof in *Shadows and residues in finite coset geometries*, and the accepted complete J_2 census, remain explicit premises. We do not repeat the subgroup census or identify historical geometry names merely by matching counts.

1 Complete inputs and archive reconciliation

The accepted census contains eight rank-two, nineteen rank-three and three rank-four geometries, with no higher-rank examples, under the stated flag-transitivity, firmness, residual connectivity, residual weak primitivity, local two-transitivity and rank-two intersection hypotheses. Write $R_9, \dots, R_{27}$ and R_{28}, R_{29}, R_{30} for its rank-three and rank-four representatives. These are identifiers in the accepted complete witness file, not the original printed geometry numbers.

The supplied `j2.rk3` archive has twenty-four active entries, after its literal final reordering; `j2.rk4` has only one. An actual ambient group isomorphism transports all these subgroup families to the accepted model of the simple group of order 604800. Matching entire families under $\mathrm{Aut}(J_2)$ and type permutations gives the following sequence for the twenty-four active rank-three entries:

$$27, 26, 25, 24, 21, 22, 23, 19, 20, 18, 17, 16, 15, 14, 14, 10, 10, 12, 12, 11, 9, 9, 13, 13.$$

Thus all nineteen classes occur; the five repetitions are $R_9, R_{10}, R_{12}, R_{13}, R_{14}$. They are not rejected geometries. The sole active rank-four entry is R_{30}; the accepted complete census supplies R_{28} and R_{29}.

For completeness, the matching tries every type permutation. For a fixed ordering it successively transports the first subgroup, restricts to its stabilizer in the automorphism group, transports the next, and continues. If one transporter exists, all such transporters form a coset of that stabilizer. Induction therefore preserves exactly all possible completions. Matching subgroup orders is only a preliminary rejection test. The final test is equality of every transported subgroup. An ambient isomorphism mapping the whole family induces a type-permuting incidence isomorphism on its cosets and all residues. Consequently every stated geometry hypothesis transfers from the accepted census. In particular, weak primitivity is retained through this exact equivalence; it is not inferred from the numerical agreement below.

2 A sparse realization of the common shadow theorem

For a finite permutation group K and subgroups $P_1, \dots, P_r$, write $P_T = \bigcap_{t \in T} P_t$ and $P_\varnothing = K$. The i-shadow of the base flag of type T is $\{P_i x : x \in P_T\}$. The original implementation realizes the whole right-coset action of degree $[K : P_i]$. The following representation avoids constructing a large universe when only small shadows are used.

Proposition 1. *For nonempty T, take a right transversal of $P_i \cap P_T$ in P_T and replace each representative x by a canonical representative of $P_i x$. This set represents the entire i-shadow*

without duplicates. Its right translate by g is obtained by replacing each representative x by the canonical representative of $P_i xg$. The empty flag may instead be represented by the cardinality $[K : P_i]$, provided its universal-set role is handled explicitly as below.

Proof. For $x, y \in P_T$, equality $P_i x = P_i y$ holds exactly when $xy^{-1} \in P_i \cap P_T$. Thus the transversal gives exactly one representative per shadow object. Canonicalization changes only its name. If $x' = hx$ with $h \in P_i$, then $P_i x' g = P_i xg$, so the translated representative is well defined and realizes the same right action. Intersections, containment, and membership are consequently unchanged.

The empty flag has the full object set as shadow. Its flag-extension identity is tautological and is counted as such. In a witness test, the computed intersection is already a subset of the i-object set. It is therefore the empty flag's shadow exactly when its cardinality is $[K : P_i]$; the empty flag is automatically incident with both tested flags. When this equality fails, the empty flag cannot witness the intersection, and every nonempty witness is tested by the displayed representation. No intersection with a fictitious empty set is performed. The same degree is used in the rank-two complete-bipartite exception. □

The normalization, double-coset enumeration and flag-constrained witness search are exactly those of the accepted common theorem. Only the representation of each finite shadow changes. Trivial panel stabilizers are allowed: the subgroup generated by an empty generator list is formed with the identity included, so a two-object panel is still recognized. The original accepted implementation remains unchanged.

A bounded comparison reconstructs both representations in 161 actual residue families whose relevant degrees are at most 2000. All 915 nonempty shadows, 6433 generator translates and 271 full-universe cardinalities agree. Six small models also agree in flag transitivity and every shadow direction, including a non-flag-transitive example; a separate trivial-stabilizer panel is checked directly. These are validation of the adapter, not a substitute for the proposition or the complete calculation on the twenty-two inputs.

3 Every rank of every residue

Theorem 2. *All nineteen rank-three inputs satisfy* $(\mathrm{IP})_2$. *Their full intersection and weak-intersection properties have the following partition:*

$$\begin{aligned} \mathrm{IP}: &\quad 9, 11, 12, 20, 24, 25, 26, 27, \\ \mathrm{WIP} \setminus \mathrm{IP}: &\quad 18, 19, 23, \\ \textit{not } \mathrm{WIP}: &\quad 10, 13, 14, 15, 16, 17, 21, 22. \end{aligned}$$

For rank four, all three inputs satisfy $(\mathrm{IP})_2$*;* R_{28} *has* $(\mathrm{IP})_3$ *and* $(\mathrm{IP})_4$*, whereas* R_{29}, R_{30} *fail both* $(\mathrm{WIP})_3$ *and* $(\mathrm{WIP})_4$*. Hence the full IP/WIP partition is one IP example and two non-weak examples.*

Proof. For every proper fixed-type set J, construct the actual residue group H_J and its remaining parabolics $H_J \cap H_i$. The accepted census and complete-family correspondence provide the required geometry hypotheses. The new run additionally checks every flag-extension equality, residual parabolic generation, panel size and two-transitivity, and both rank-two shadow directions; it does not repeat the weak-primitivity computation. For each remaining target type it exhausts the normalized object–flag orbits and all admissible witness flags. Every negative decision stores an actual normalizing permutation and the failed shadow intersection.

There are 178 base residues including panels, of which 109 have rank at least two. Their 255 directional decisions are retained in full. For each rank k, $(\mathrm{IP})_k$ requires all directions in all base residues; $(\mathrm{WIP})_k$ allows a different successful direction in each residue. Actual residue

multiplicities are $[G : H_J]$. The saved complete decisions and these quantifiers give the asserted partition. Full IP and WIP take the conjunction over ranks $2 \leq k \leq$ rank; rank one is automatic and rank zero is not an existential weak test. □

The aggregate columns are therefore $(8, 3, 8)$ in rank three and $(1, 0, 2)$ in rank four, agreeing with the corresponding rows of Tables 1, 3 and 7. Tables 2, 4 and 8 use identifiers from the earlier classification. The present correspondence does not establish those historical names; in particular it does not rename R_{28} as the printed 4.2 without a separate source identification.

The source orbit algorithm and published shadow results are attributed to Jacobs and Leemans. Catalogue completeness, its subgroup-data trust boundary, and the full accepted residue proof are retained. The sparse representation and its checks do not establish a new catalogue or a complete Lean formalization. Other groups and the full original remain outside this bounded packet. There is no new source, question or numerical credit. Source Section 6 contains executable Magma; any later measurement must treat concrete source and replacement programs symmetrically.